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***B.Tech. Degree I Semester Regular and Supplementary/
II Semester Supplementary Examination November 2018***

**CE/EE/ME/SE GE 15-1105 A & CS/EC/IT GE 15-1205 B
BASIC MECHANICAL ENGINEERING
(2015 Scheme)**

Time : 3 Hours

Maximum Marks : 60

PART A
(Answer *ALL* questions)

(10 × 2 = 20)

- I. (a) Name and explain the law that forms the basis for temperature measurement.
- (b) Differentiate between boiler mountings and accessories. List their names.
- (c) How are IC engines classified? Also mention the applications of IC engines.
- (d) Describe the mechanism of a simple vapour compression refrigeration system.
- (e) Give the range of specific speeds for Kaplan, Francis and Pelton wheel turbines.
- (f) List the merits and demerits of diesel engine power plants.
- (g) Explain the contributions of Gilberth to the management theory.
- (h) Describe various welding defects and their remedies.
- (i) Explain the procedures followed while casting a component.
- (j) Briefly explain different types of gears.



PART B

(4 × 10 = 40)

- II. (a) Define internal energy. Is it a function of a state or process? (2)
 - (b) An insulated cylinder of 0.4 m diameter and 0.8 m length contains 10 kg of oxygen. Paddle work is done on the gas to increase its pressure from 3 bar to 6 bar. Determine the change in internal energy, work done on the gas and the change in enthalpy. (8)
 $C_p = 0.91 \text{ kJ/kgK}$ and $C_v = 0.64 \text{ kJ/kgK}$.
- OR**
- III. (a) Describe the various losses occurring in steam turbines. (3)
 - (b) What is the necessity of compounding a steam turbine? Discuss the salient features of the various methods used for compounding of steam turbines. (7)
- IV. (a) State the objectives of lubrication in an IC engine. (3)
 - (b) The following particulars refer to a 4 cylinder, 4 stroke petrol engine. (7)

Bore	=	65 mm
Stroke length	=	95 mm
Torque developed	=	64 Nm when engine turns 3000 rev/min.
Clearance volume	=	63 cm ³ for each cylinder.
Relative efficiency	=	0.5
Calorific value of petrol	=	42000 kJ/kg.

 Calculate brake mean effective pressure and the fuel consumption in kg/hr.

OR

(P.T.O.)

- V. (a) What is meant by coefficient of performance? Explain. (2)
 (b) With neat sketches explain summer and winter air conditioning systems. (8)

- VI. (a) Differentiate between a turbine and a pump. (3)
 (b) The following data is related to a Pelton wheel turbine : (7)

Head at the base of the nozzle	=	80 m
Diameter of the jet	=	100 mm
Discharge of the nozzle	=	$0.30 \text{ m}^3/\text{s}$
Power at the shaft	=	206 kW
Power absorbed in mechanical resistance	=	4.5 kW

- Determine (i) Power lost in nozzle.
 (ii) Power lost due to hydraulic resistance in the runner.

OR

- VII. (a) State the essential elements of a hydro electric power plant. (4)
 (b) Define productivity. What are the various means of improving productivity? (6)

- VIII. (a) Differentiate between brazing and soldering. (2)
 (b) Describe the principle of resistance welding process and explain different types of resistance welding processes. (8)

OR

- IX. (a) Derive an expression for the ratio of belt tensions in a flat belt drive. (4)
 (b) Power is transmitted between two shafts by a V-belt whose mass is 0.9 kg/m length. The maximum permissible tension in the belt is limited to 2.2 kN . The angle of lap is 170° and the groove angle 40° . If the coefficient of friction between the belt and pulleys is 0.17 , find : (6)
 (i) Velocity of the belt for maximum power.
 (ii) Power transmitted at this velocity.
